`Voice City`.

The user would not merely choose among prerecorded voices. They would **design a synthetic narrator** by adjusting a large multidimensional control surface, previewing the result, and saving it as a reusable voice model.

The key architectural decision

We should not literally expose hundreds of independent neural-network parameters. That would be unpredictable and nearly impossible for users to control.

Instead, the interface can present hundreds of meaningful dials while translating them into a smaller, validated set of:

- speaker-identity embeddings
- prosody controls
- acoustic-style embeddings
- pronunciation rules
- emotion curves
- text-dependent performance instructions
- deterministic random seeds

Modern generative speech research supports separating speaker identity, vocal style, emotion, rhythm, pauses, and intonation rather than treating a voice as one fixed recording. (arXiv)

Voice city experience

1. Voice DNA

This section defines the fundamental physical character of the voice.

Controls would include:

Control family	Example dials
Perceived age	youthful, mature, elderly
Vocal weight	light, medium, heavy
Pitch center	low to high
Pitch range	narrow to theatrical

Resonance	chest, throat, mouth, nasal, head
Brightness	dark to brilliant
Texture	clean, airy, breathy, raspy, gravelly
Vocal-fold character	soft, firm, creaky, pressed
Body size impression	small to imposing
Gender presentation	continuously adjustable rather than categorical
Warmth	clinical to intimate
Presence	distant to commanding
Articulation	soft to highly precise
Timbre complexity	pure to harmonically rich
Uniqueness	conventional to unusual

These controls generate a **base speaker embedding** representing a completely synthetic identity.

2. Performance Console

This controls how the voice acts while narrating.

- speaking rate
- sentence rhythm
- phrase length
- pause frequency
- pause duration
- emphasis strength
- pitch movement
- melodic variation
- energy
- emotional intensity
- restraint
- confidence
- intimacy
- authority
- friendliness
- suspense

- humor
- sadness
- excitement
- urgency
- conversationality
- theatricality
- breath frequency
- vocal effort
- ending cadence
- question inflection
- paragraph-reset behavior

A voice could therefore remain recognizable while having multiple performances:

- **Mara — Neutral Narration**
- **Mara — Intimate Memoir**
- **Mara — Suspense**
- **Mara — Energetic Advertisement**

3. Accent and speech behavior

The user could define:

- language
- regional accent
- accent strength
- dialect vocabulary
- rhoticity
- vowel placement
- consonant sharpness
- syllable reduction
- formality
- code-switching behavior
- foreign-word pronunciation
- proper-name pronunciation
- number and date reading
- acronym handling

Accent controls must be constrained to prevent unstable or caricature-like results.

4. Advanced Soundboard

An expert mode can expose the “hundreds of dials” concept.

Rather than displaying everything at once, controls would be grouped into expandable modules:

1. Source and vocal folds
2. Formants and resonance
3. Pitch and melody
4. Timing and rhythm
5. Articulation
6. Breath and texture
7. Emotion
8. Accent and phonology
9. Narration behavior
10. Recording environment
11. Post-processing
12. Text interpretation

Users could switch between:

- **Simple:** approximately 12 macro controls
- **Studio:** approximately 50 controls
- **Laboratory:** 150 to 300 controls
- **Automation:** editable curves over the audiobook timeline

5. Randomize and mutate

This would make voice creation accessible even to non-engineers.

Generate

The user supplies a description such as:

> Mature, reassuring female narrator with a low center pitch, restrained emotion, warm resonance, precise diction, and subtle British influence.

The system produces several synthetic candidates.

Mutate

The user selects a voice and requests:

- warmer
- 15 percent older
- less nasal
- more authoritative
- slower
- slightly rougher

- wider emotional range
- retain identity but reduce brightness

Breed voices

Two synthetic voices could be blended:

- 70 percent Voice A
- 30 percent Voice B

This must apply only to synthetic or properly licensed source voices.

Lock characteristics

The user could lock selected characteristics before randomizing:

- lock timbre
- lock age
- lock accent
- randomize performance
- randomize resonance only

6. Audition room

Every change should support near-real-time preview.

The user selects standardized audition scripts for:

- fiction
- nonfiction
- dialogue
- technical writing
- legal material
- emotional passages
- advertisements
- children’s narration
- difficult names
- numbers and acronyms

The UI would offer:

- A/B comparison
- blind comparison
- waveform comparison
- loudness-matched playback

- sentence-by-sentence comparison
- save candidate
- reject candidate
- compare up to eight versions

7. Saved Voice Model

A created voice becomes a versioned asset:

```None
Voice
  Identity model
  Base parameter set
  Speaker embedding
  Default performance profile
  Pronunciation dictionary
  Supported languages
  Model version
  Random seed
  Training model version
  Safety classification
  Ownership and licensing record
```

A saved voice would have:

- stable `voice_id`
- user-visible name
- organization ownership
- version history
- preview samples
- tags
- default use cases
- supported languages
- quality score
- model compatibility
- creation provenance
- export restrictions
- deletion and revocation controls

Critical distinction: voice preset versus voice model

The product should support both.

Voice preset

A parameter recipe applied to an existing underlying model.

Advantages:

- inexpensive
- almost instant
- easy to version
- no dedicated training
- ideal for MVP

Synthetic voice model

A persistent speaker identity generated in latent space and optimized for consistency across long-form narration.

Advantages:

- stronger identity
- better stability across books
- more distinctive results
- can support specialized pronunciation and performance behavior

This may require an asynchronous optimization job, GPU inference infrastructure, model storage, and model-version compatibility controls.

Recommended technical architecture

New provider abstraction

The current provider interface accepts text, a `voice_id`, and an engine. Voice city needs a richer contract.

```python
class GenerativeVoiceProvider(ABC):
    def create_voice(
        self,
        identity_parameters: dict,
```

```
        style_parameters: dict,
        seed: int,
    ) -> VoiceArtifact:
        ...

    def preview_voice(
        self,
        voice_artifact_id: str,
        text: str,
        performance_parameters: dict,
    ) -> bytes:
        ...

    def synthesize(
        self,
        text: str,
        voice_artifact_id: str,
        performance_parameters: dict,
    ) -> bytes:
        ...
```

The existing Edge and Polly providers can remain available as traditional catalog providers. Voice city would be a new provider class rather than forcing every provider to support voice generation.

New backend services

```
None
backend/services/voice_city/
    parameter_schema.py
    parameter_mapper.py
    voice_generator.py
    voice_optimizer.py
    preview_renderer.py
    embedding_store.py
    safety_classifier.py
```

```
pronunciation_engine.py
preset_service.py
version_service.py
```

Proposed database entities

`voices`

- `id`
- `organization_id`
- `created_by`
- `name`
- `description`
- `voice_type`
- `status`
- `provider`
- `model_family`
- `current_version_id`
- `visibility`
- `created_at`
- `updated_at`
- `deleted_at`

`voice_versions`

- `id`
- `voice_id`
- `version_number`
- `identity_parameters`
- `default_style_parameters`
- `speaker_embedding_key`
- `model_artifact_key`
- `model_revision`
- `seed`
- `quality_score`
- `consistency_score`

- `supported_languages`
- `status`
- `created_at`

`voice_presets`

- `id`
- `organization_id`
- `name`
- `parameters`
- `source_voice_version_id`
- `category`
- `is_template`

`voice_previews`

- `id`
- `voice_version_id`
- `text`
- `parameter_overrides`
- `audio_key`
- `duration`
- `created_at`

`voice_generation_jobs`

- `id`
- `organization_id`
- `voice_id`
- `status`
- `progress`
- `stage`
- `error`
- `locked_by`
- `locked_at`
- `attempts`
- `created_at`

The existing durable Postgres job system should be reused for these jobs rather than introducing a second queue.

Parameter representation

Each voice version should store an immutable canonical JSON document:


```json
{
  "schema_version": "1.0",
  "identity": {
    "pitch_center": -0.28,
    "pitch_range": 0.42,
    "vocal_weight": 0.63,
    "brightness": -0.15,
    "breathiness": 0.18,
    "roughness": 0.09,
    "warmth": 0.74,
    "resonance": {
      "chest": 0.62,
      "mouth": 0.31,
      "nasal": 0.07
    }
  },
  "performance": {
    "rate": 0.92,
    "energy": 0.44,
    "expressiveness": 0.36,
    "pause_density": 0.58,
    "emphasis": 0.31
  },
  "accent": {
    "locale": "en-US",
    "region": "mid-atlantic",
    "strength": 0.16
  },
  "seed": 481928
}
```


The saved JSON is essential for:

- deterministic regeneration
- version comparison
- rollback
- API use
- reproducible audiobook revisions
- migrating a voice between future model versions

UI structure

New navigation item

Voices → Voice city

Main screen

None

```
┌ Voice city ─────────────────────────────────────────┐
│ Voice: Eleanor V4       [Save Version] [Generate Variants]│
├────────────────┬─────────────────────────┬─────────────────┤
│ Voice DNA      │ Live Audition           │ Candidate Tray  │
│                │                         │                 │
│ Age        ●── │ “It was nearly…”        │ A  ▶  Save      │
│ Weight     ─●─ │                         │ B  ▶  Compare   │
│ Warmth    ──●─ │  ▶ Play                 │ C  ▶  Reject    │
│ Texture    ●── │  A/B                     │                 │
│ Pitch      ─●─ │                         │                 │
│ Resonance  ──● │ Waveform                │                 │
├────────────────┴─────────────────────────┴─────────────────┤
│ Performance | Accent | Emotion | Pronunciation | Advanced │
└────────────────────────────────────────────────────────────┘
```

Soundboard behaviors

- every adjustment is undoable
- controls can be searched by name
- related controls highlight together
- impossible combinations are automatically constrained
- hovering explains audible impact

- double-click resets a dial
- users can type exact values
- snapshots can be compared
- MIDI controllers may eventually manipulate controls physically
- parameter automation can vary by chapter, character, scene, or sentence

Safety and ownership

The safest initial version is **synthetic voice creation without reference audio**.

That means the system generates new voices from parameter space rather than copying a named person. Reference-based cloning can be a separate, heavily controlled workflow.

Required protections include:

- block prompts requesting imitation of named public or private individuals
- compare generated embeddings against protected or prohibited voice registries
- require consent for any uploaded reference recording
- maintain provenance for every generated voice
- watermark or fingerprint generated audio
- log creation and export activity
- allow organization administrators to revoke a voice
- prevent anonymous public model exports
- define prohibited impersonation use
- support takedown and incident investigation

Recommended delivery sequence

Phase 1: Parametric presets

Build approximately 40 meaningful controls over an open or internally hosted controllable TTS model.

Deliver:

- Voice city UI
- live preview
- deterministic seeds
- presets
- saved voice versions
- audiobook synthesis using saved voices

- no cloning
- no per-user model training

Phase 2: High-dimensional Voice DNA

Add:

- 150-plus advanced controls
- latent interpolation
- voice mutation
- random generation
- characteristic locking
- candidate comparison
- quality and consistency scoring

Phase 3: Persistent synthetic identities

Add:

- voice optimization jobs
- persistent embeddings
- long-form identity consistency testing
- multilingual variants
- audiobook-specific fine-tuning
- pronunciation adaptation

Phase 4: Character and scene direction

Add:

- multiple voices per book
- automatic dialogue detection
- character casting
- chapter-level styles
- emotion curves
- director instructions
- timeline automation

Phase 5: Authorized reference voices

Only after the synthetic system is mature:

- consented reference uploads
- talent agreements
- identity verification
- speaker authorization
- revocation
- auditing
- protected-person similarity checks

Product requirement

A voice should be considered production-ready only when it passes:

- identity consistency across at least 30 minutes
- pronunciation accuracy
- chapter-to-chapter timbre stability
- emotional controllability
- no unexpected speaker drift
- loudness consistency
- long-form listening evaluation
- duplicate/similarity safety screening

The right product is therefore not simply “hundreds of EQ knobs.” It is a **visual interface over a controlled latent speech model**, giving ordinary users simple macros and sound designers access to the deeper voice parameter space. This should become one of ACX City’s defining capabilities.